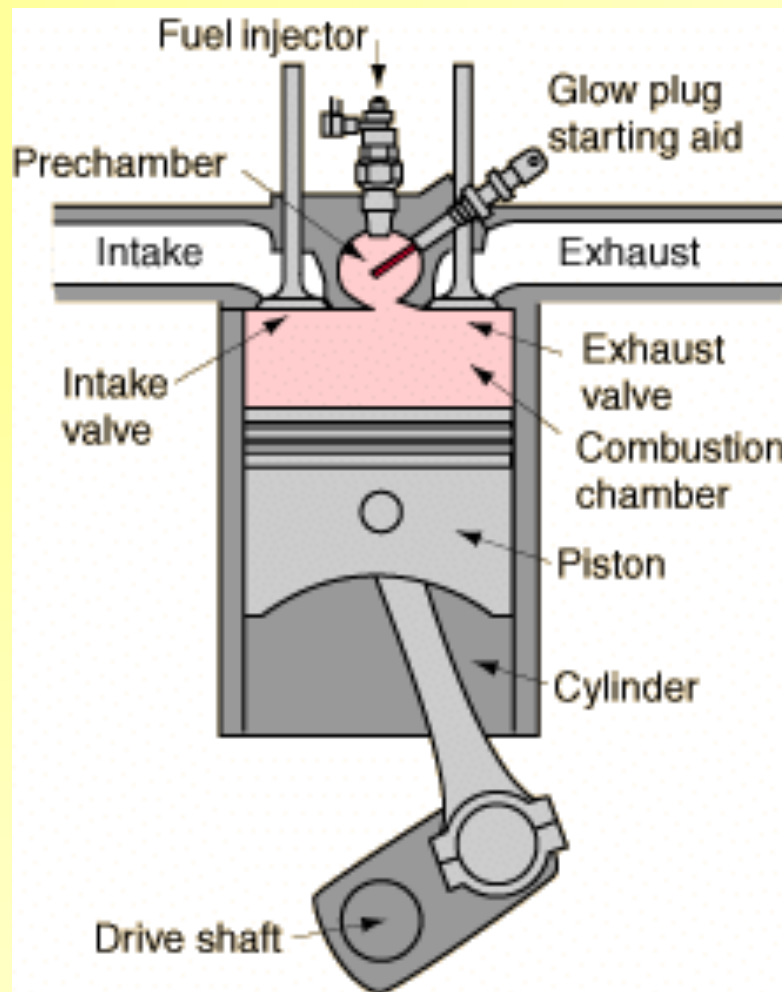




COMPRESSION IGNITION ENGINES



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SYLLABUS



S.I. Engines: Deviation from the ideal cycle, combustion losses, flame development and propagation, effects of operating variable on detonation; Detonation rating of engine fuels, HUCR Dope, preignition combustion chambers, Mechanics of carburetion, MPFI systems, mixture requirement and combustion.

C.I. Engines: Stages of combustion, Diesel knock, effect of operating variables, knock rating of C.I. engine fuels, Cetane number; Effect of additives, combustion chambers. Requirements of fuel injection system and its mechanism, Fuel filters.

Lubrication and Cooling: Forced, splash, by-pass and direct, Blow by and crankcase dilution.

Air and water cooling, Thermostatic control.

Performance and Testing: High speed indicators, various efficiencies, Morse Test and Willan's line; torque and mean effective pressure, performance and heat balance sheet, Measurement of volumetric efficiency, Effect of atmospheric condition on performance of I.C. Engines; Supercharging; high altitude problem of I.C. Engines, methods and types of supercharging; suitability of S.I. and C.I. Engines;

Air Pollution: Vehicular emission, its norms, Causes and control of air pollution from diesel and petrol engines, catalytic converter.

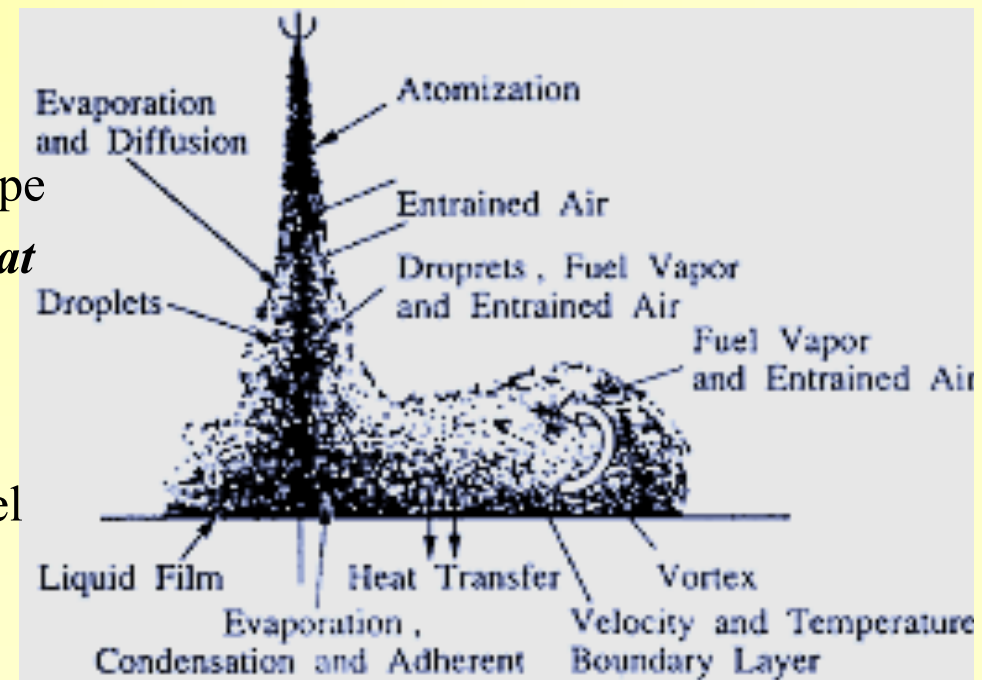
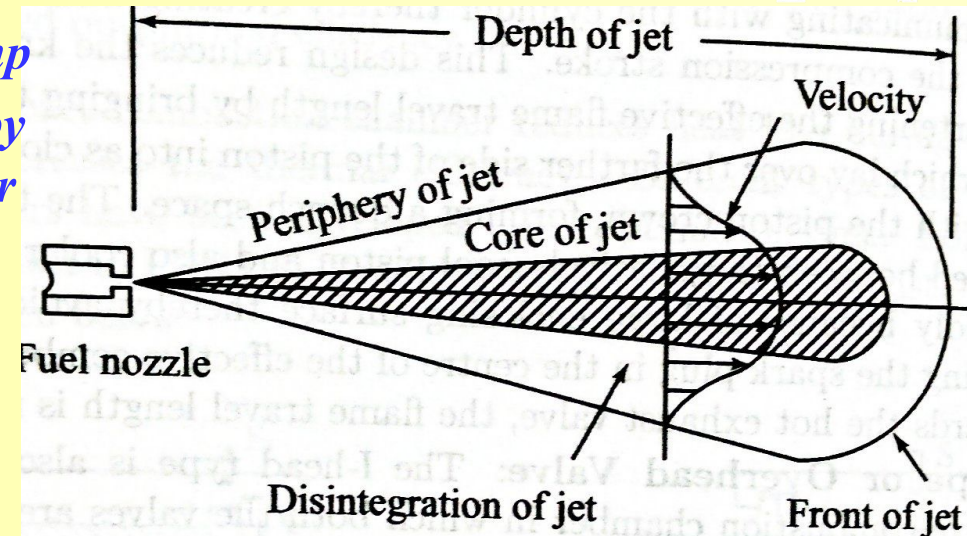
Introduction to dual fuel and multi-fuel engines; free-piston engine and rotary combustion engine.



FUEL COMBUSTION IN CI ENGINES

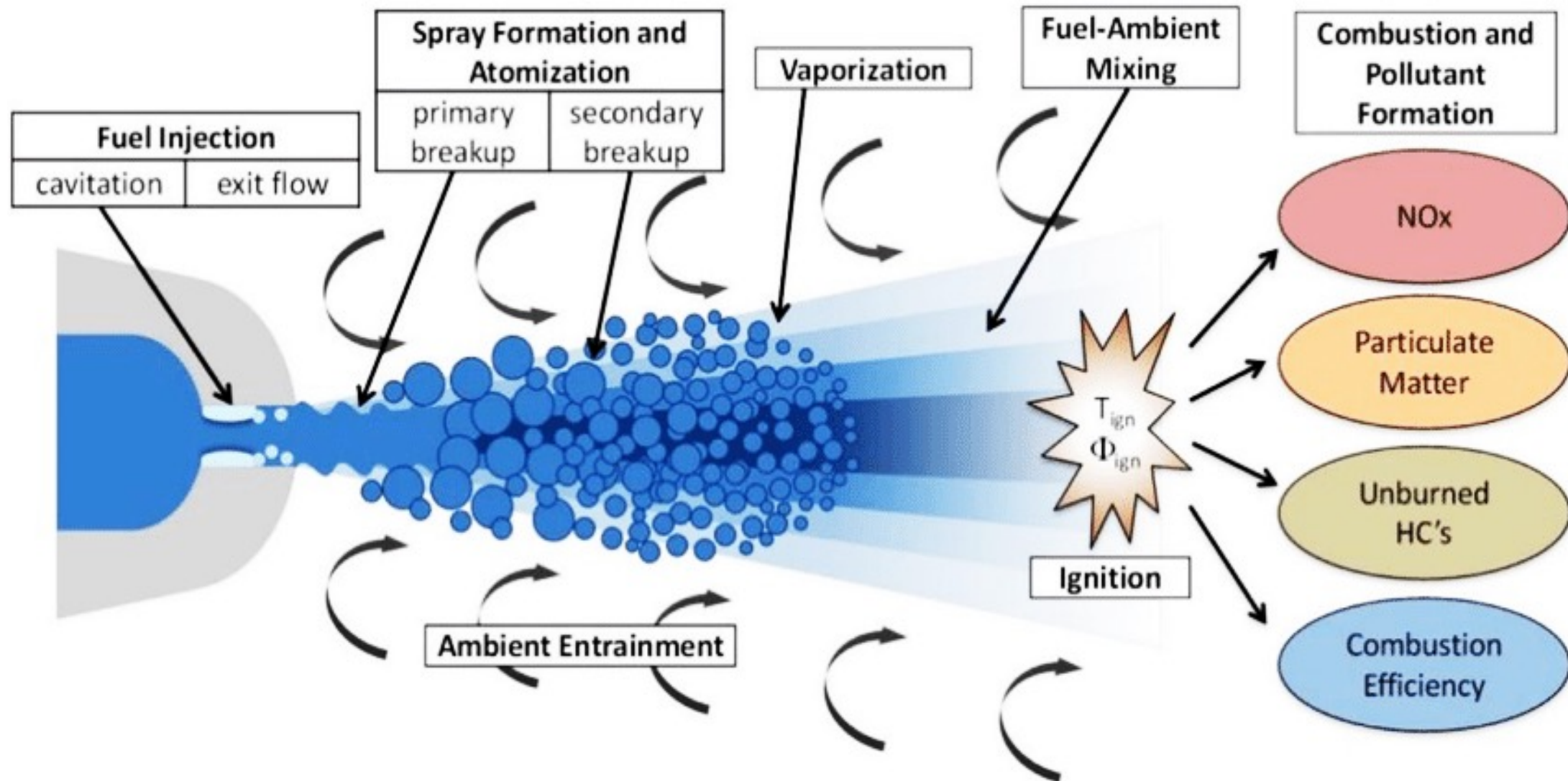


- Due to high CR ($\sim 12-22$) air compressed, rise in pr-temp
- Diesel injected by single/multi-jets ($\sim 20^\circ - 35^\circ$ bTDC) by high pr pump (~ 110 to 200 bar) in combustion chamber
- At injection, Fuel jet disintegrates as
 - ✓ The core of spray jet is *surrounded* by
 - ✓ i. *Air envelope* and ii. *Fuel particles*.
 - ✓ Fuel *Atomization & Vaporization* create a spray envelope.
 - ✓ Air *turbulence* in CC pass through jet, tears fuel particles from core, results in better mixing of A-F
 - ✓ Thus, proper *Air-Fuel* mixtur made in spray envelope
 - ✓ The, fuel droplets *evaporate* by absorbing *latent heat* of *vaporization* from the surrounding *air*,
 - *Droplets reduce temp of air and some time elapsed before rise in temperature & oxidation/combustion*
 - ✓ Vapour temp becomes higher than *Fire Point* of fuel
 - ✓ As air-fuel vapour reaches to ignition temp in combustible range, *auto-ignition* take place & fuel starts combustion, consume air-fuel mixture.





COMBUSTION PHENOMENON IN CI ENGINES





STAGES OF COMBUSTION IN CI ENGINES

➤ To burn heterogenous mix, air produce swirl, combustion start at many points in jet.

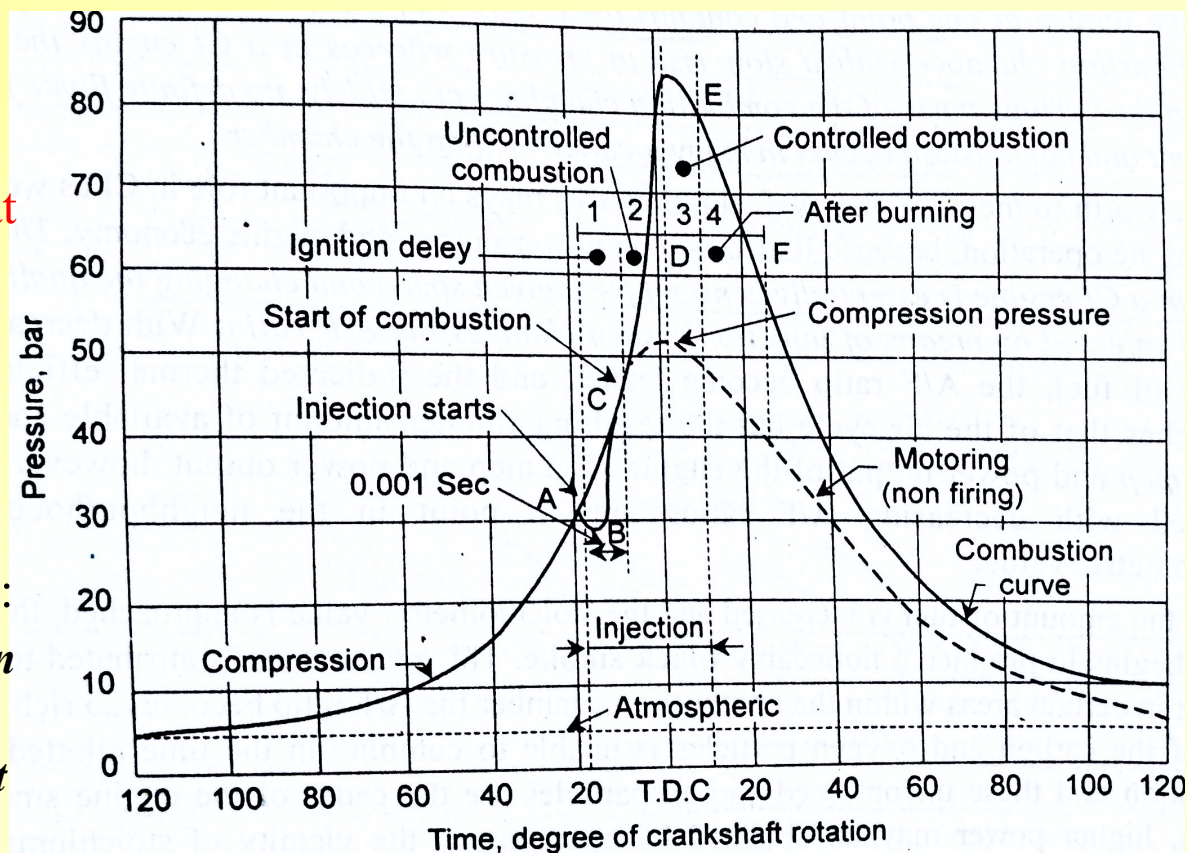
- ✓ At compression, fuel spray at 'A', due to atomization-vaporization jet disintegrates & core surrounded by A+F envelope,
- ✓ Attain self-ignition temp & pre-chemical reaction starts before actual-burning.

➤ Stages of Fuel combustion in CI Engines

- ✓ **1. Ignition delay:** Fuel vapourizes by heat gain from compd-air cause pr drop 'A-B'
- ✓ After vapourization, **pre-flame** reaction start at B, energy liberate & pr rise 'B-C'
- ✓ A definite **period of inactivity** betwn fuel injection at 'A' & start of actual burning at 'C' called delay period. It is of 2-types:
 - **Physical delay:** time spent in atomization vapourization, mixing A-F & temp rise
 - **Chemical delay:** pre-flame reaction start at 'B', slow burning during B-C, flame propagates and consume the charge.

Stages of Combustion in CI Engine

1. Ignition Delay
2. Rapid or Uncontrolled Combustion
3. Controlled Combustion
4. Afterburning





STAGES OF COMBUSTION IN CI ENGINES ...



➤ 2. Rapid/uncontrolled combustion, 'CD'

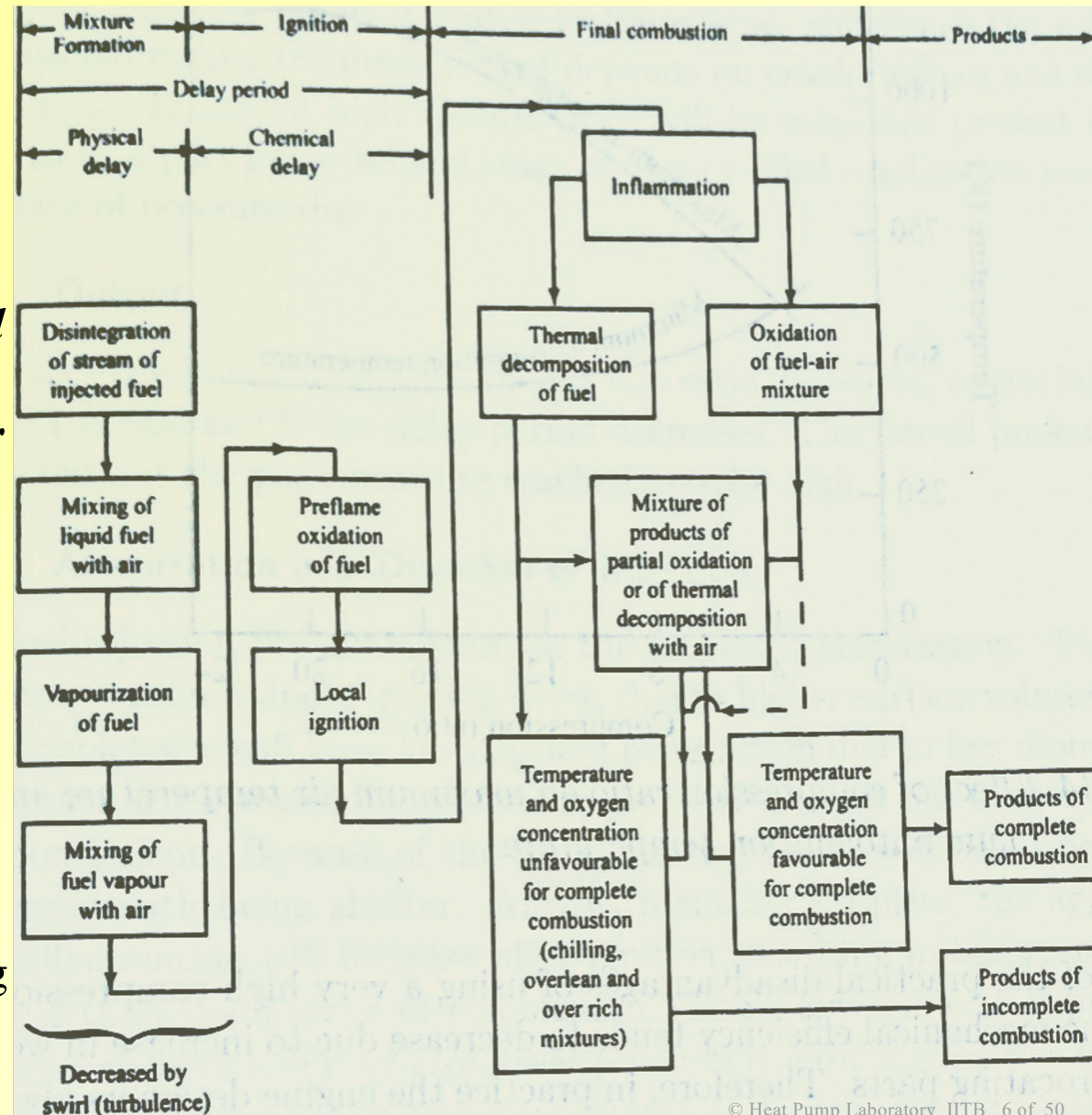
- ✓ At end of delay period, ignition **start** in fuel envelope, **pre-flame** completed, **flame propagate** through A+F charge where **concentration** optimum (0-100).
- ✓ **Period** of rapid combustion is from **end of delay** period/beginning of combustion to the point of **max pressur rise** in indicator diagram 'C-D'.

➤ 3. Controlled Combustion, 'DE': max p-T

- ✓ Charge in II-stage burn faster with reduced ignition delay, finds necessary O_2 & further rise in pr is controlled by fuel **injection** at end of combustion.

➤ 4. After Burning, 'EF'

- ✓ Combustion is incomplete upto TDC point, the **rest fuel** particles burn during the **expansion** process i.e. **burning** at **end** of injection cause smokey exhaust.





DETONATION IN CI ENGINES

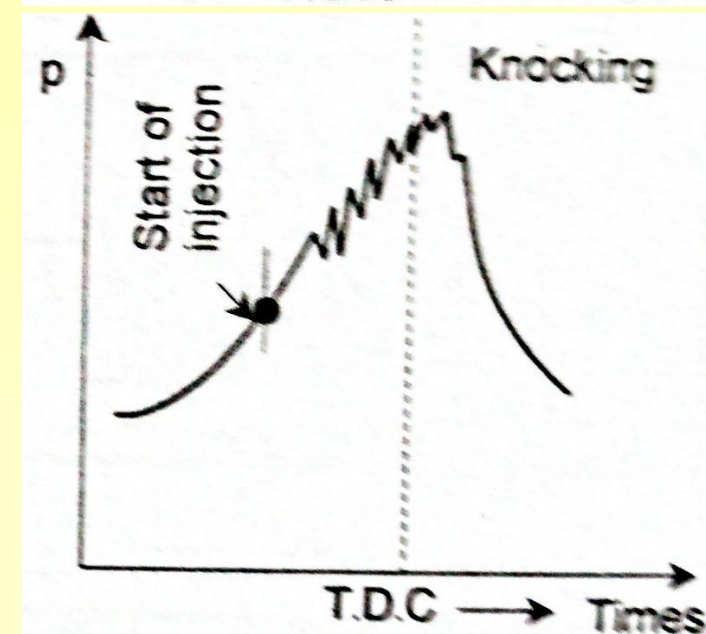
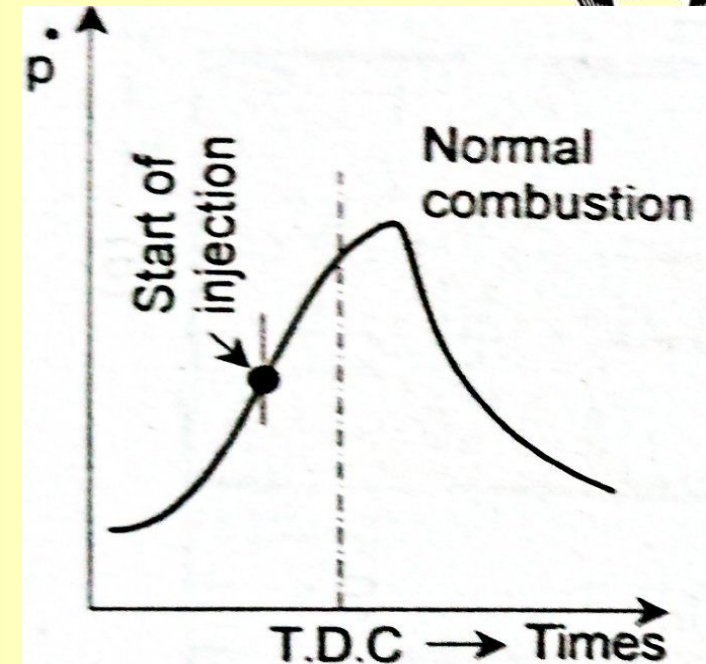


➤ Normal Combustion

- ✓ The fuel injection in CI engines takes place for a certain *time interval*, droplets passing through the *ignition delay* & *additional* droplets injected in CC during expansion.
- ✓ In case of *shorter* delay period, the *first droplets* of fuel will commence *actual* 'burning' in short time span after injection & a *small* amount of droplets will accumulate.
- ✓ Consequently, the burning of mixture will produce a *Smooth* pressure rise & combustion is said *Normal*, Fig 1.
- ✓ If delay period is *short*, combustion is *Smooth, orderly* & in *sequential controlled* manner, is *Normal* combustion.

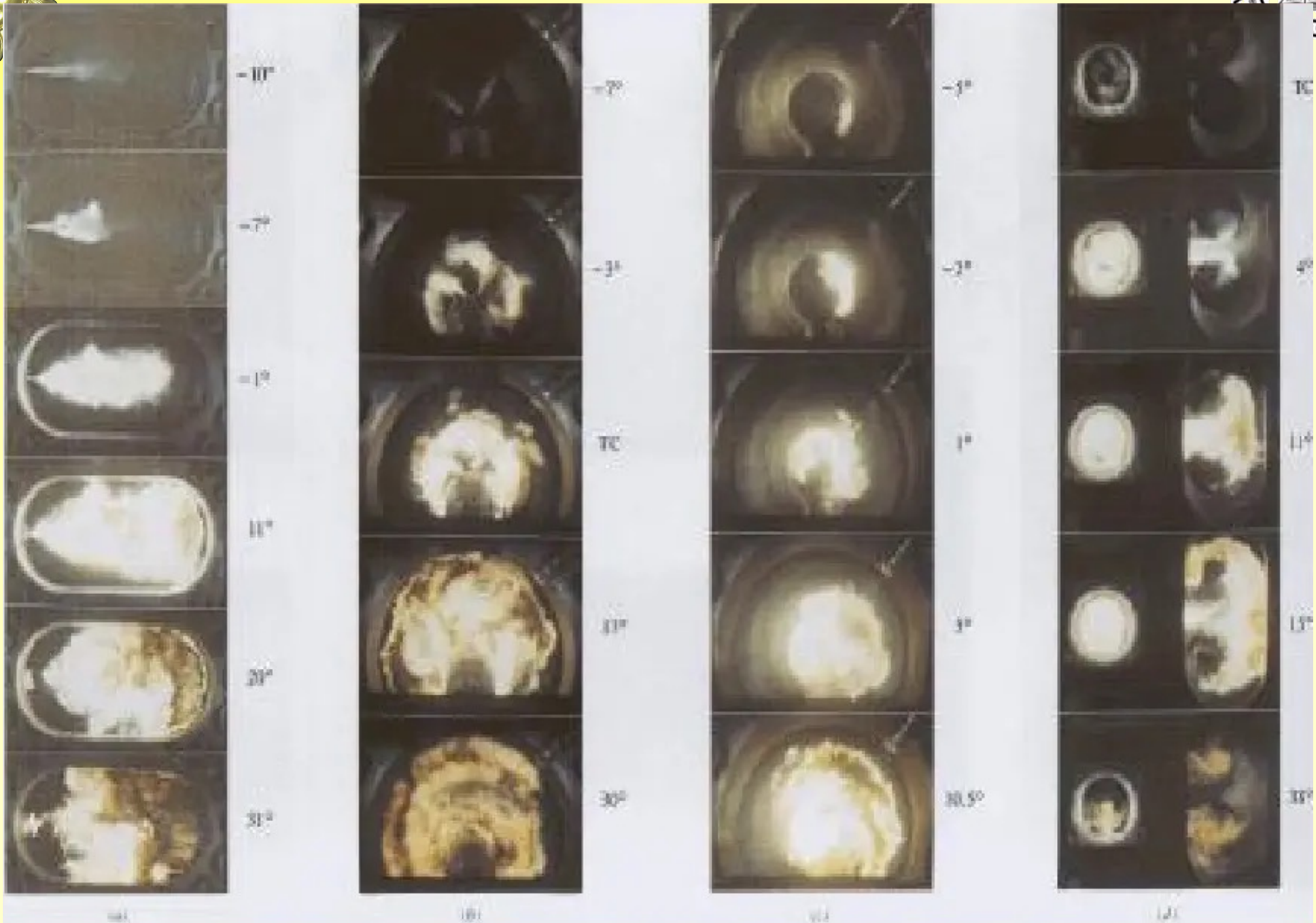
➤ Abnormal Combustion/ Detonation:

- ✓ If ignition delay period is *longer*, then actual burning of *first droplets* is delayed and *large quantity* of fuel-droplets will be *accumulated* in the engine combustion chamber.
- ✓ When *actual* burning starts, the ignition of *accumulated* fuel cause a *violent* and *instantaneous* rise of pr, as Fig 2..
- ✓ In such conditions *extreme* pr differentials are produced and violent gas *vibrations* occur, known as *Detonation*.





STAGES OF DETONATION

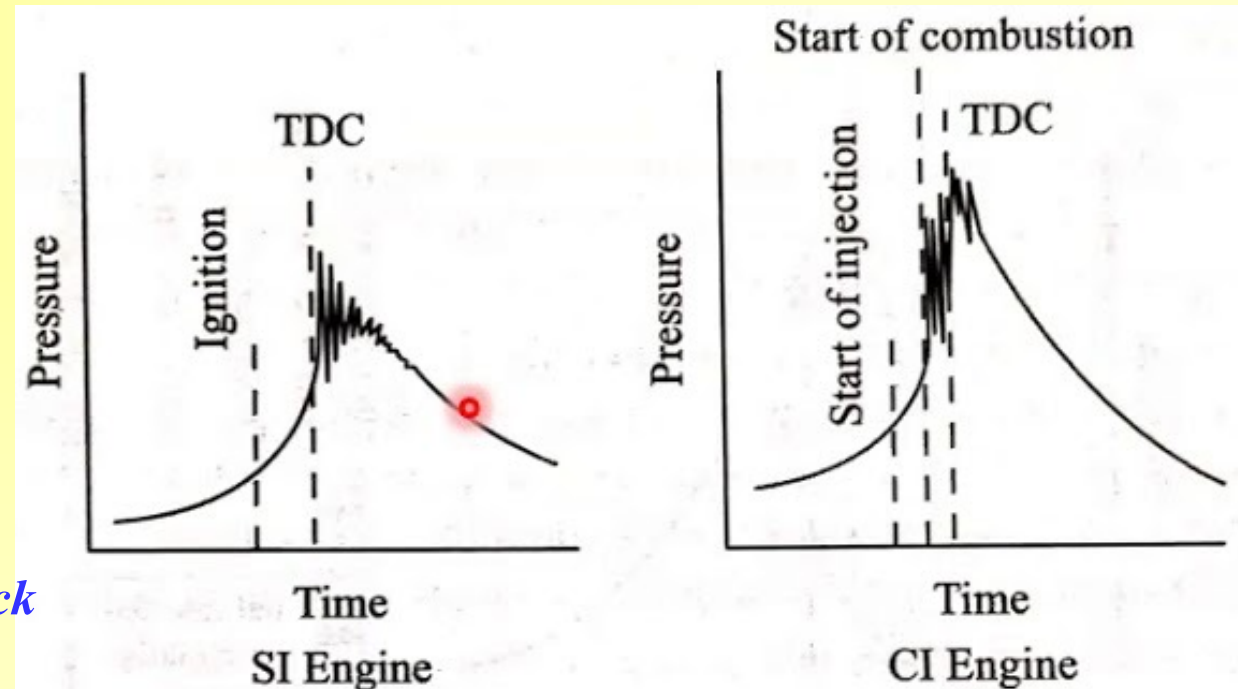




KNOCKING / DETONATION

- In SI/CI engines knocking is fundamentally due to the Auto-ignition of the fuel-air mixture.
- In both cases the knocking depends on the auto ignition-lag/delay period of fuel-air mixture
- **Parameters**

- ✓ Fuel quality: Superior/Inferior fuels
- ✓ Combustion chamber design
- ✓ Intake condition (p , T , ρ) of air
- ✓ Fuel-self ignition temperature



- If controlled above factors, avoided knock
- **Methods of Knock-control**

- ✓ Use better **cetane rating** fuel: Cetane Number (**45 to 65**) has lower delay period reduces knock
- ✓ Controlling the rate of **fuel supply**: by injecting less fuel at beginning and more amount later in combustion chamber by use of definite cam profile.
- ✓ Knock reduction by use of **multiple jets/injector** to enhance swirl/turbulence in chamber.
- ✓ By use of **ignition accelerators**: CN can be increased by adding chemicals call **dopes** – ethyl nitrate and amyl-nitrate in concentration of 8.8 gm/litre and 7.7 gm/litre, reduce knock.



EFFECT OF OPERATING VARIABLES



- *Combustion process in CI engines is controlled by length of delay period. If delay period is short, engine runs smoothly. Longer delay period makes engine run in rough & noisier rise to knock.*
- ✓ *Inlet temp*: higher inlet air temp increases in cylinder *temperature* at injection, *decreases* delay period, decreases the knock tendency.
 - ✓ *Inlet pressure*: high inlet pr result in high compression pressure, *reduces* the knock tendency
 - ✓ *Compression ratio*: high CR brings high-pressure, temperature of air at *injection*, results *short* delay period and normal pressure rise, reduces knocking tendency.
 - ✓ *Injection timing*: if fuel injection is *too early* bTDC, delay period *longer* due to low pressure & temperature of air, resulted in abnormal pressure rise, *favourable* for knocking.
 - ✓ *Engine speed*: at high speed more fuel injected during delay period and mixing of air fuel mixture will be better, both causes high rate of pressure rise giving *rise* to knocking.
 - ✓ *Engine size*: large engines run at low speed hence quantity of fuel injected during delay period is less so compression temp is higher, these effects *reduce* the knock tendency.
 - ✓ *Coolant temp*: increase in jacket temp, decreases delay period, *reduces* the knock tendency.
 - ✓ *Injection pressure*: increase in injection pressure, decreases delay period due to high degree of atomization, *reduces* the knock tendency.
 - ✓ *Type of CC*: pre-combustion gives a shorter delay period compare to open, *reduces* the knock.
 - ✓ *Turbulence*: it promotes homogenous mixture, *decreases* delay period & knock tendency.



EFFECTS OF OPERATING VARIABLES ON DETONATION IN CI ENGINES



- ✓ **Volatility**: homogeneous volatile fuels with *low* viscosity, *reduce* physical delay & knock
- ✓ **Chemical composition**: fuels of *high* self ignition temp, *longer* delay periods, increases knock
- ✓ **Additives**: the ethyl nitrate, amyl-nitrate etc. *reduce* ignition delay period, so knock tendency
- ✓ **Load**: increase in load F/A, increases operating temp as delay period *decreases*, reduce knock.
- ✓ **Swirl rate**: change in swirl changes fuel evaporation, mixing, affect heat transfer, *reduces* the delay period and knock
- ✓ **Oxygen** concentration: decreases delay period, so *reduces* knock
- ✓ **Knock** characteristics: good fuel should have short ignition lag and will ignite more readily.
- ✓ **Starting** characteristics: easy starting, high cetane: volatile mixture, self-ignition temp low
- ✓ **Smoke & odour**: fuel should not promote smoke or order in exhaust, complete combustion.
- ✓ **Handling** ease: the fuel should have a high flashpoint and a high fire point.

➤ *From above discussion, it is obvious that to reduce the possibility of knock in CI engines, the First Element of fuel and air should have*

- ✓ 1. High-temperature
- ✓ 2. High-density
- ✓ 3. Short delay period and
- ✓ 4. Reactive mixture



FUELS OF CI ENGINES

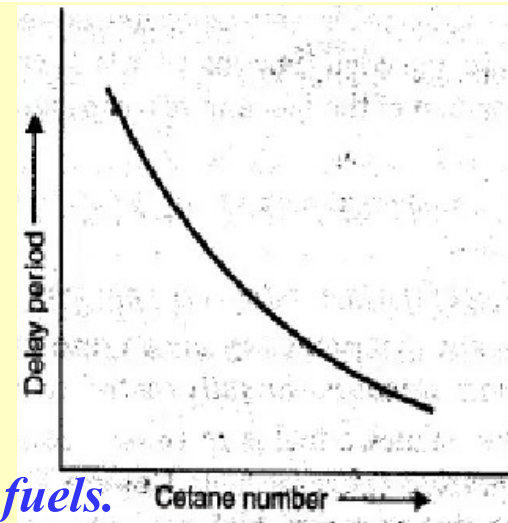


➤ *The CI engine fuels are obtained from fractions of crude petroleum. The diesel fuels are heavier and more viscous than gasoline. Following are the main characteristics of CI fuels:*

- ✓ **Ignition lag:** it should have short ignition delay, measured in terms of **cetane number**
- ✓ **Volatility:** good for mixing and combustion, have sufficient volatility at operating temp range
- ✓ **Viscosity:** should be **low** to reduce the knock tendency.
- ✓ **Impurities:** to prevent excessive wear, impurities must be **reduced**.
- ✓ **Ash content:** ash content of the burnt fuel should be **minimum**.
- ✓ **Pour point:** the fuel should have an **optimum** value of pour point.
- ✓ **Flashpoint:** should be **optimum** for safety and fire hazards.
- ✓ **Gravity:** measure of density, sp gravity is the ratio of density of substance to the density of water.
- ✓ **Ignition quality:** good ignition quality means **short delay period**, measured by cetane number.
 - **Cold starting:** high cetane number fuels is recommended.
 - **Compression issue:** efficiency increases with the CR should be minimum for the starting.
 - **Engine roughness:** signifies to withstand against the intensity of vibrations cause due to knock.
 - **Corrosion and wear:** should not give rise to corrosion and wear before and after combustion.
 - **Cloud point:** temperature below which the wax content of fuel separates out in the form of a solid,
 - **Cleanly Ness:** dirt is undesirable in diesel fuel, cause rapid wear, high degree of filtration is required.



CI FUEL RATINGS



- *The knock resistance in CI engines depend on*
 - ✓ Chemical characteristics of fuel and
 - ✓ Design & operating conditions of engine.
- *Knock rating of diesel fuel is found by comparing it with primary reference fuels.*
- *CI engine fuels are rated by their Cetane number. The property of ignition lag/delay period is measured in terms of Cetane number. Higher the Cetane rating of fuel, lesser is the propensity for diesel knock. The Cetane number of a diesel fuel is a measure of its ignition quality.*
- *The Cetane number of a fuel is the % (by volume) of Normal-Cetane in a mixture of*
 - ✓ Normal Cetane ($C_{16}H_{34}$) is arbitrarily assigned a Cetane number of **100** and
 - ✓ α -Methyl Naphthalene ($C_{11}H_{10}$) with a Cetane number of **0**.
 - ✓ The percentage by volume of normal Cetane in a mixture of normal Cetane and α -Methyl Naphthalene which has the same ignition characteristics/delay period as the test fuel when combustion is carried out in a standard engine under specified operating conditions.
- *Cetane rating is the measure of best fuel, if have cetane rating sufficiently high to avoid knock.*
- *Knock resistance: a property of diesel fuel can be improved by adding small quantities of dopes/compounds like amyl-nitrate, ethyl-nitrate or ether.*
- *Good SI engine fuel is bad CI fuel, high octane number implies low Cetane number and vice versa. The straight chain paraffins are the most suitable fuel for CI engines.*

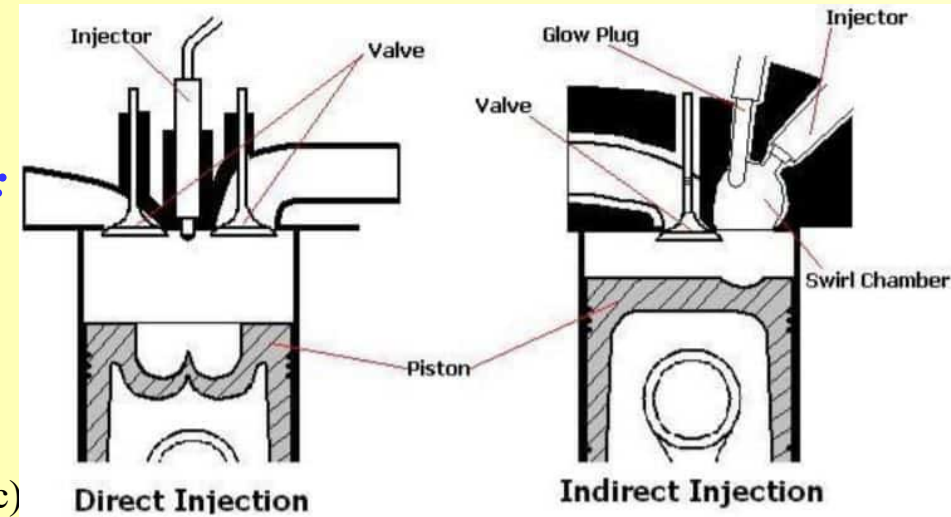


COMBUSTION CHAMBERS

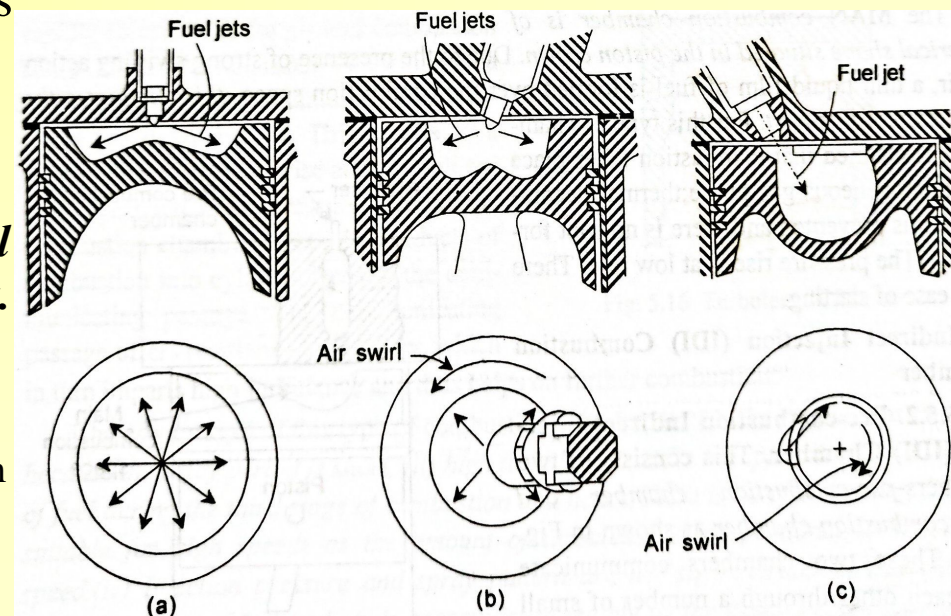


- **Function:** Generate air-swirl / turbulence for better mixing in controlled manner and short time.
- **Combustion Chambers can be divided in two categories:**

- ✓ 1. **Direct injection** : open type CC (bowl-in-piston)
 - Fuel injected directly in upper part of cyl act as CC
 - Improve air motion by (i) shrouding inlet valve-low speed (ii) squish rapid air movement walls to center
 - ❖ (a) shallow depth-large engines (b) Hemispherical CC (c) toroidal cone 150–160° (d) cylindrical with side injector



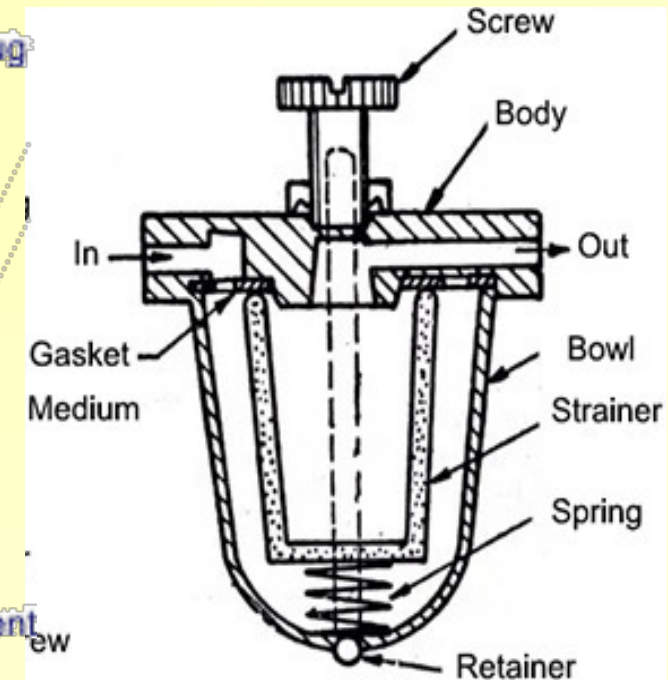
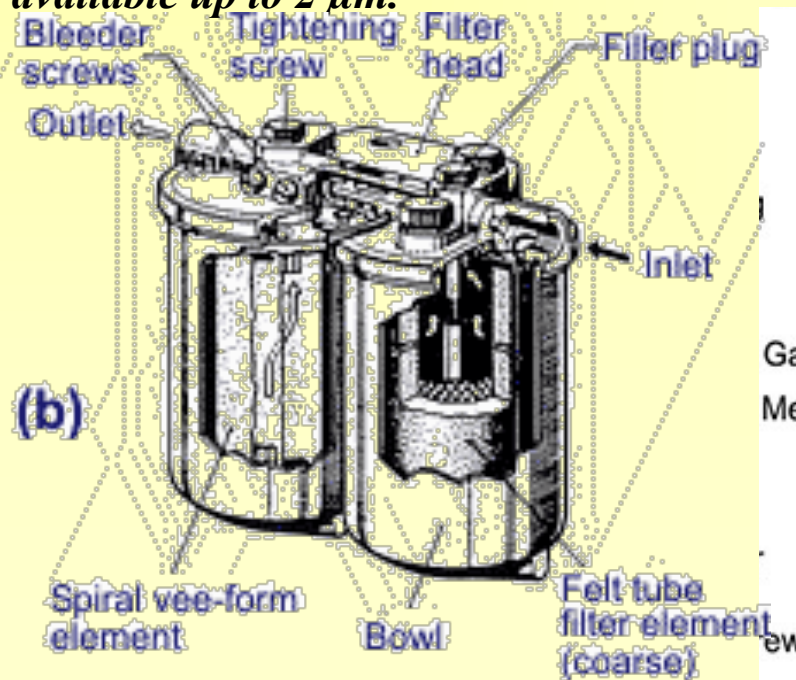
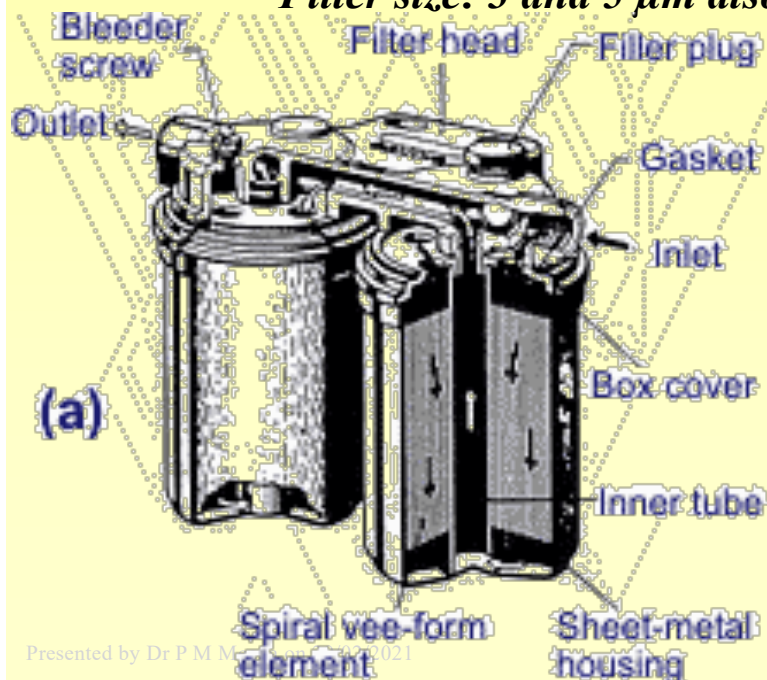
- ✓ 2. **Indirect injection**: CC divided in two regions /CCs
 - a. **Pre-combustion** chamber connected to b. **Main** combustion C via narrow passage nozzle / orifices
 - Fuel injected in pre-combustion chamber have ~1/3 of total charge, at combustion induces very high swirl in main CC, helps in reduction of delay period/knock.
 - A very high swirl created in main CC due to 90-95% gases pushed out of pre-CC, oxidizes. Main-CCs are,
 - ❖ a. Turbulent / swirl type combustion chambers, in which very high swirl is created during compression stroke.
 - ❖ b. Air cell or small CC (Lonva & Aero-type) in which both compression and combustion swirls introduced.





DIESEL FUEL FILTERS

- *The purpose of fuel filter in diesel engine is to remove foreign particles from fuel, have primary and secondary fuel filters to ensure supply of cleaned fuel before it enters in engine. It filters out,*
 - ✓ Capable of removing dirt particles, harmful impurities & moisture to
 - *Reduce wear-tear, avoid clogging & improve life of diesel engines.*
- *Diesel filters are of two types*
 - ✓ Primary filter: First filter located between fuel tank and supply pump
 - *Cleans out large particles sized 10-15 μm down to 30 μm also separates water from fuel.*
 - ✓ Secondary filter: filters the fine particles and located between supply pump and injection pump,
 - *Filter size: 3 and 5 μm also available up to 2 μm .*





REQUIREMENTS OF FUEL INJECTION SYSTEM

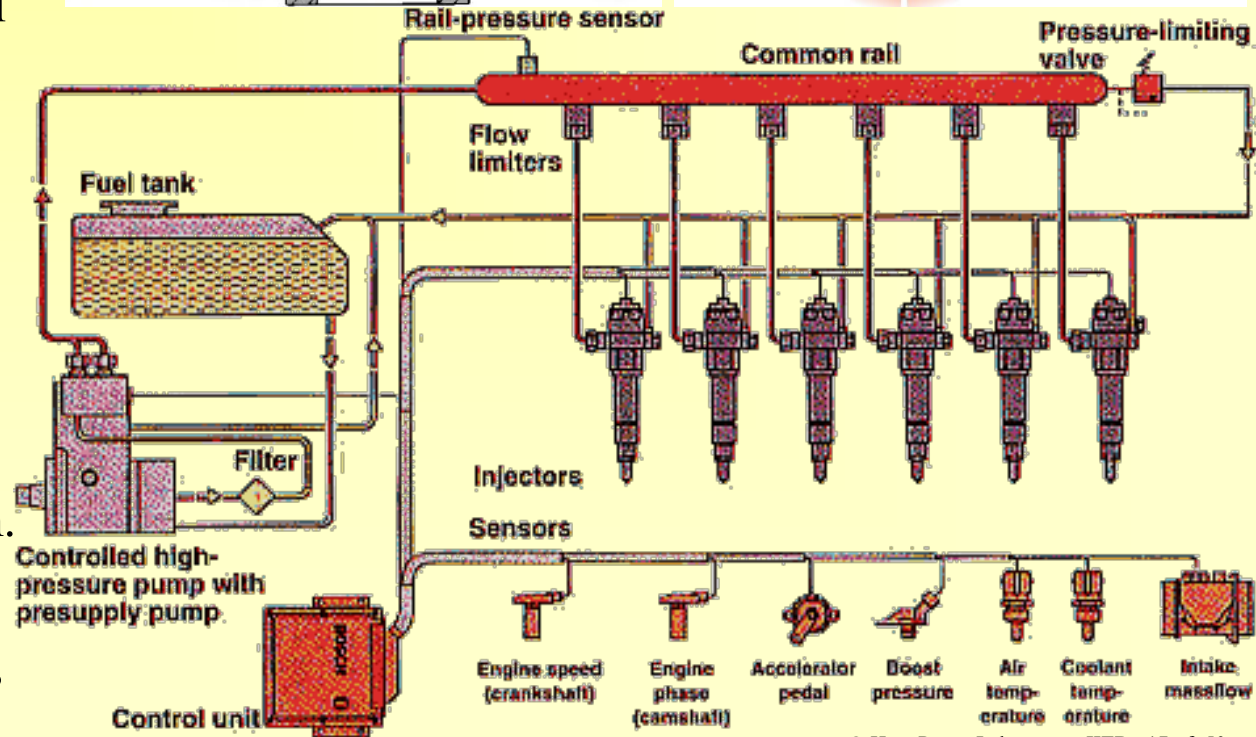
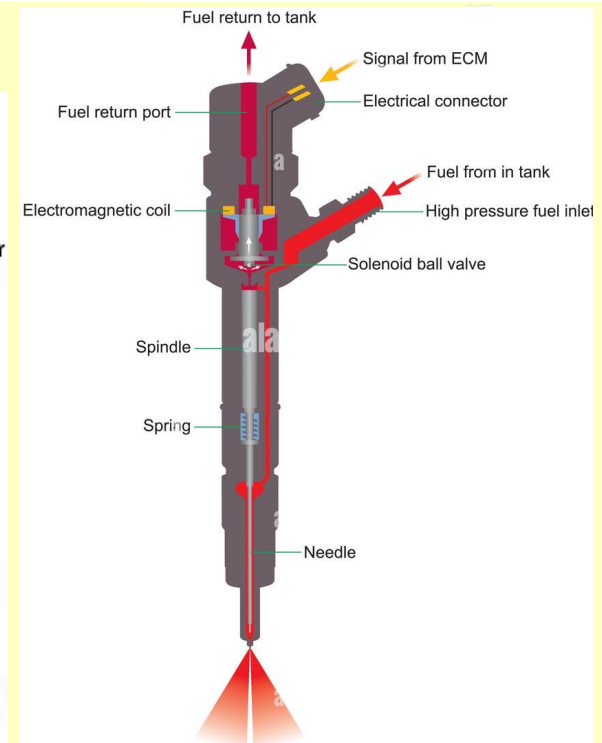
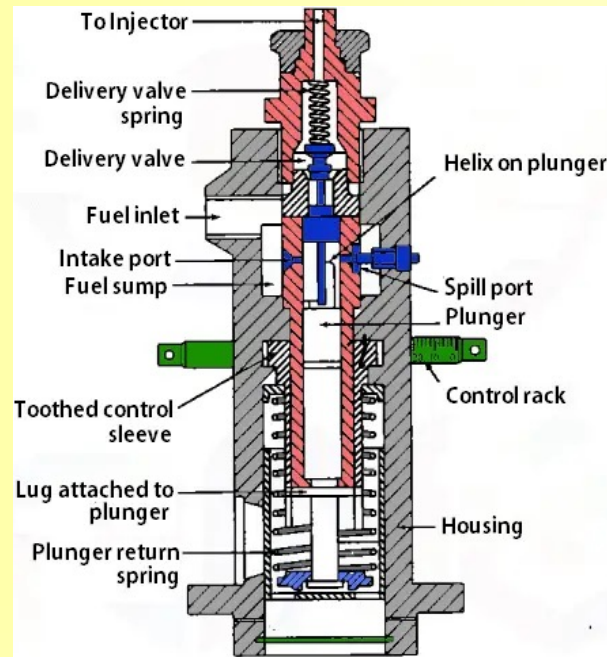


- *Major requirements of a fuel injection system: to affectively make use of fuel injection system,*
 - ✓ **Pumping Elements:** to supply fuel from fuel tank to the engine cylinder.
 - ✓ **Metering Elements:** (accurate fuel metering per cycle and supply as per engine load & speed)
The correct amount of fuel must be delivered to meet power requirement. $Q = a.V.t.n$
 - ✓ **Timing Control** (to adjust the start and end of injection process in the cycle): Fuel must be injected at proper time to obtain maximum power ensuring fuel economy and clean burning.
 - ✓ **Mixing Element:** (to atomate and distribute the fuel into very fine droplets throughout the combustion chamber) Proper spray pattern to ensure rapid mixing of air and fuel mixture.
 - ✓ **Distribution Elements:** to deliver the metered fuel in equal quantities among the multi-cylinder engines. Uniform distribution of fuel-droplets combustion chamber
 - ✓ **Prevention** of impingement of fuel on the cylinder walls.
 - ✓ **No lag** between beginning & end of injection i.e. eliminate dribbling of fuel droplets into cylinder
- *Advantages:*
 - ✓ Better atomization & distribution of fuel, higher BMEP, allows inferior fuels combustion
- *Disadvantages:*
 - ✓ It requires complicated mechanism to run the compressor
 - ✓ The weight of the engine increases.
 - ✓ Part of the engine power is used to drive the compressor, so BHP of engine is reduced.



FUEL INJECTION SYSTEMS

- *Engine performance: power, economic operation etc. dependent on fuel system*
- *Injection system perform the duty of fuel injection, combustion initiation & control the flame travelling*
- *Components of fuel injection system:*
 - ✓ **Fuel tank:** fuel stored in main tank
 - ✓ **Fuel pump:** meters & pressurize fuel from tank to injector via supply line
 - ✓ **Governor:** supply fuel accordance to changing conditions of loads/speed
 - ✓ **Injector:** atomize the fuel in engine cylinders
 - ✓ **Fuel filters:** to remove dust and oppressive particles from fuel which may damage injection/engine system.
- *Fuel can be Injected by*
 - ✓ Individual pump single/multi-jet inj,
 - ✓ Common rail direct injection system

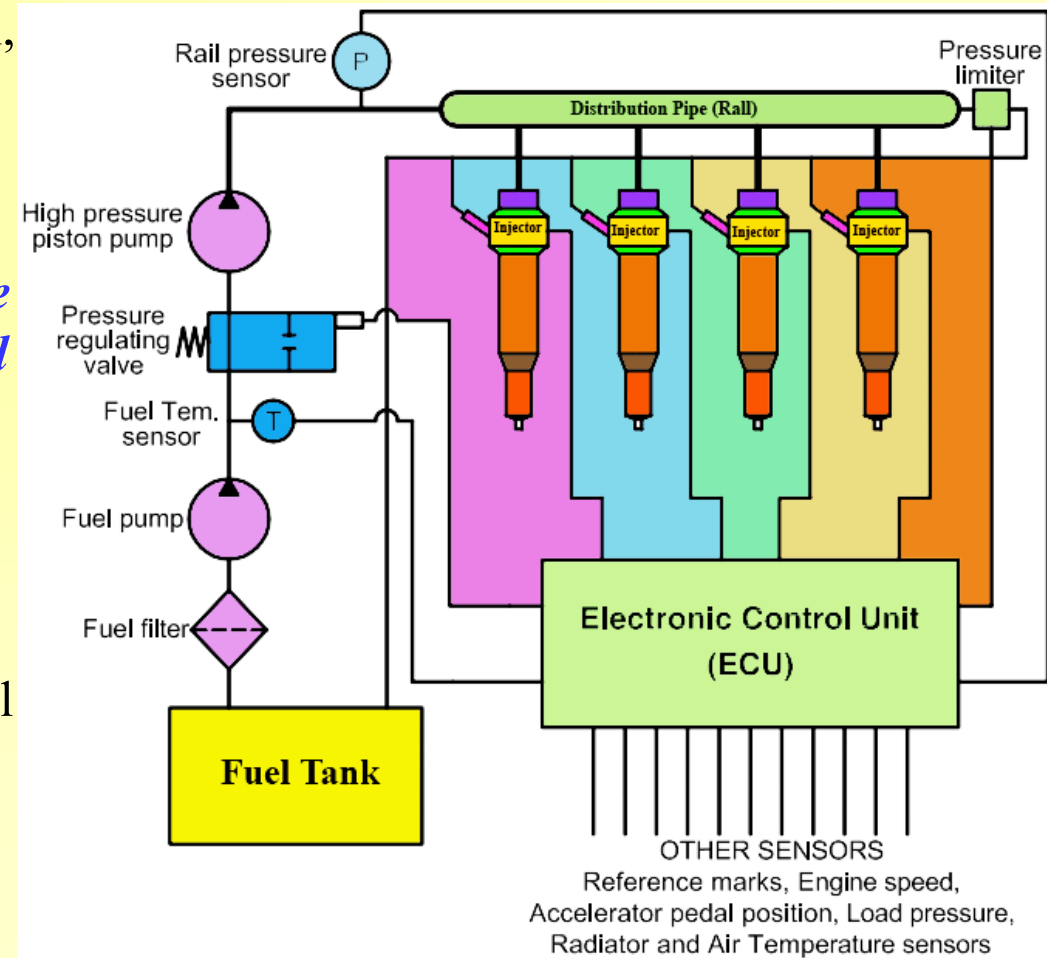




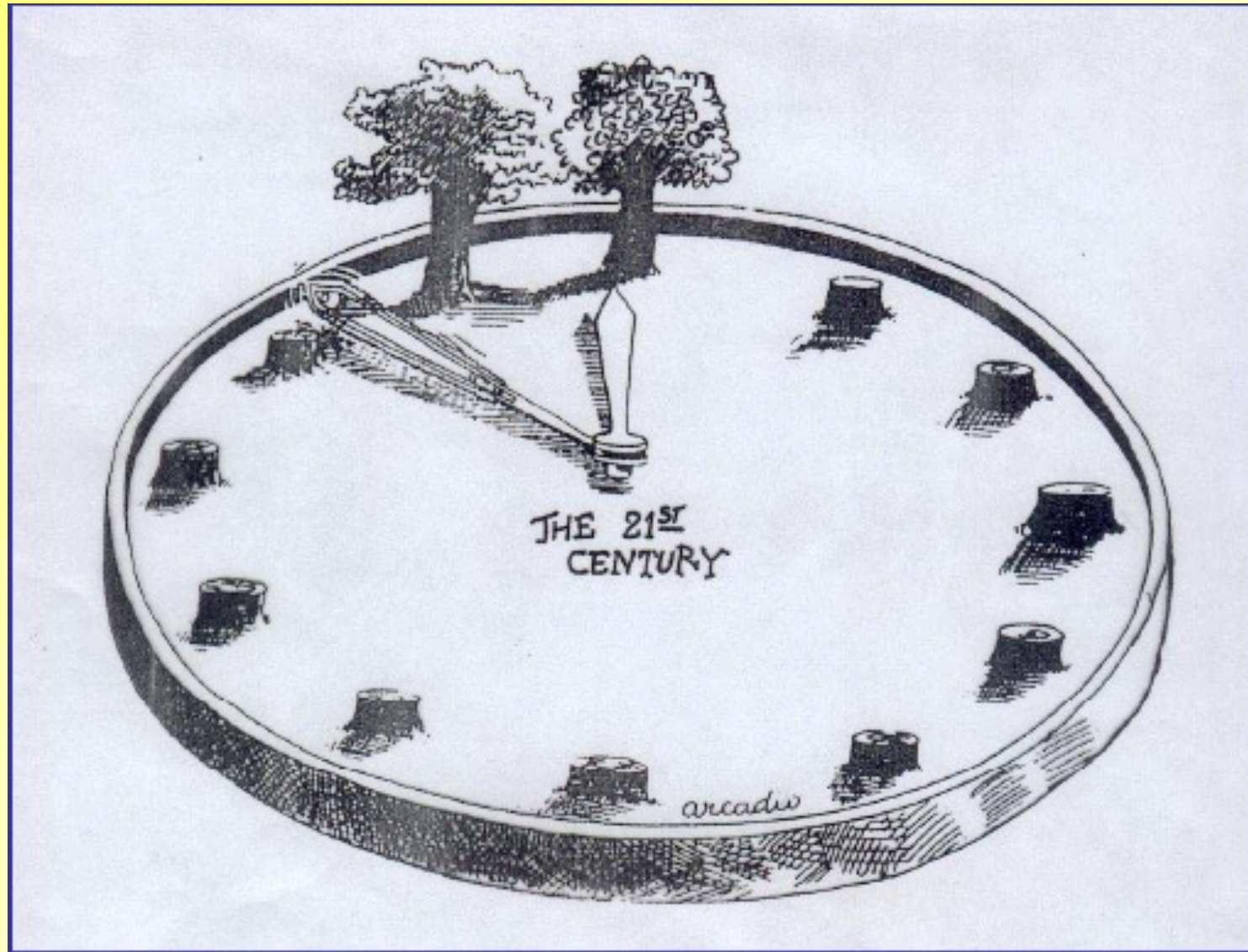
CRDI FUEL INJECTION SYSTEM



- *A regular diesel engine uses low-pr fuel pump while CRDI houses a high-pr pump controlled by ECU like turbocharger originated in aircrafts.*
 - ✓ Key components of CRDI: common rail, injector and ECU (Electronic Control Unit) are.
 - ✓ The pump takes diesel from tank and stores in common rail at a very high pressure.
- *Dynamic Fuel Injection :ECU takes care of engine RPM, temp, air density, power etc; delivers optimal amount of fuel accordance to changing conditions*
 - ✓ The common rail holds pressurized fuel,
 - ✓ Injector inject-atomize the fuel in fine droplets
 - ✓ Engine has a lot of smaller droplets in cylinder.
- *Optimized Combustion:* small droplets of diesel fuel then explode with the help of compression
- *Advantages:*
 - ✓ Better Performance, Enhanced Fuel Efficiency,
 - ✓ Refinement, Engine Durability,
 - ✓ Lower Emissions: Air, water & Noise pollution



Common Rail Direct Injection (CRDI) System



Thank You